Supplementary Material

Do topological and quantum-inspired molecular representations add value?

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1 Topological persistence and resistance resilience

This supplementary analysis tests whether the topological features introduced in the main manuscript carry an association with computational resistance-resilience scores (RRS) beyond molecular size and related structural covariates. It includes $n = 494$ compounds with complete H_1 and RRS profiles; the analysis is exploratory and is not used to claim an independent topological biomarker.

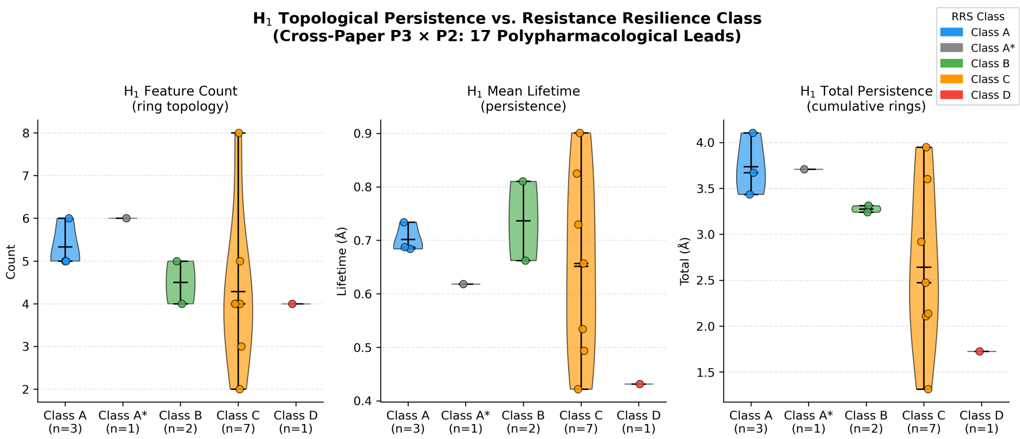

Figure S1: *H_1–RRS analysis. The violin plot shows the exploratory subset; class-level differences are not interpreted as independent effects. In the expanded cohort ($n = 494$), H_1 count showed a weak unadjusted association with continuous RRS (Spearman $\rho = 0.2399$, $p = 6.76 \times 10^{-8}$), but the partial association was null after controlling for molecular weight, ring count, fraction sp^3, and H_0 count ($\rho_{partial} = 0.0329$, $p = 0.4668$). The result is size-mediated and hypothesis-generating, not evidence of an independent topological predictor.*

2 Pharmacophore fingerprint representation

The two-dimensional pharmacophore (PHCO) baseline uses RDKit's `SparseBitVect` representation. Active bit positions were transferred explicitly to the numerical feature matrix before cross-validation. Under this representation, PHCO achieved an AUC of 0.896 on the 19 836-molecule complete-embedding panel (Table 1).

3 Per-fold activity-prediction estimates

Table S1 gives per-fold AUC, accuracy, and F1 for six representation methods. ECFP4, TFP, and TNE use the 19 836-molecule complete-embedding panel; the hybrid uses the corresponding full analysis panel; QKS and RBF use a 5 000-molecule 6-qubit quantum-kernel panel. Summary estimates are given in the main manuscript (Table 1).

Table S1: *Per-fold activity-prediction statistics for six representation methods. ECFP4, TFP, and TNE use the 19 836-molecule Random Forest benchmark; the hybrid uses the corresponding Random Forest analysis; QKS and RBF use the 5 000-molecule 6-qubit state-vector SVM analysis. Because the rows differ in population and classifier, they are presented as complementary summaries rather than as a direct cross-method comparison. Summary estimates from the primary benchmark are given in the main manuscript (Table 1).*

Method	Fold	AUC	Accuracy	F1
ECFP4	1	0.947	0.898	0.933
ECFP4	2	0.950	0.894	0.930
ECFP4	3	0.948	0.895	0.930
ECFP4	4	0.952	0.899	0.934
ECFP4	5	0.940	0.893	0.930
Hybrid (TFP+TNE+QK)	1	0.894	0.848	0.902
Hybrid (TFP+TNE+QK)	2	0.887	0.844	0.900
Hybrid (TFP+TNE+QK)	3	0.886	0.844	0.900
Hybrid (TFP+TNE+QK)	4	0.893	0.846	0.902
Hybrid (TFP+TNE+QK)	5	0.878	0.836	0.895
TFP (TDA)	1	0.881	0.846	0.902
TFP (TDA)	2	0.875	0.836	0.895
TFP (TDA)	3	0.880	0.846	0.902
TFP (TDA)	4	0.878	0.838	0.897
TFP (TDA)	5	0.866	0.826	0.889
TNE ($d = 8$)	1	0.720	0.753	0.856
TNE ($d = 8$)	2	0.722	0.756	0.857
TNE ($d = 8$)	3	0.733	0.755	0.857
TNE ($d = 8$)	4	0.721	0.758	0.858
TNE ($d = 8$)	5	0.714	0.760	0.859
QKS (quantum, 6q)	1	0.791	0.813	0.881
QKS (quantum, 6q)	2	0.843	0.815	0.881
QKS (quantum, 6q)	3	0.833	0.826	0.890
QKS (quantum, 6q)	4	0.800	0.818	0.887
QKS (quantum, 6q)	5	0.833	0.841	0.900
RBF (gamma-tuned)	1	0.812	0.821	0.886
RBF (gamma-tuned)	2	0.839	0.815	0.881
RBF (gamma-tuned)	3	0.844	0.833	0.894
RBF (gamma-tuned)	4	0.818	0.827	0.892
RBF (gamma-tuned)	5	0.818	0.843	0.900

Complete per-fold records are included in the supplementary data archive.

4 Interpretation and cross-study scope

The TNE docking analysis measures information retention on the molecular set used to derive the embeddings; it does not assess prospective binding to new compounds. TNE was competitive with ECFP4 for PfDHFR (TNE $R^2 = 0.473$ versus ECFP4 $R^2 = 0.461$) but lower for PfATP4 and PfCRT; these target-specific results are therefore descriptive and do not establish a universal advantage. The expanded H_1–RRS analysis is hypothesis-generating: the unadjusted association is attenuated after adjustment for molecular size and related structural covariates, so no independent topological predictor is claimed. The four-complex P2 MD cohort is distinct from the P3 representation panel, and the present analysis does not provide experimental activity measurements.

5 Topological fingerprint statistics

Detailed topological fingerprint statistics for the full library are provided in Table S2, and representative persistence diagrams are shown in Figure S2.

Table S2: *Topological fingerprint statistics across 19 849 valid molecules (Vietoris–Rips persistent homology, three homology dimensions).*

Feature	Mean	Std	Min	Max
H_0 entropy	3.5781	0.2014	2.2887	4.1943
H_0 count	38.0417	7.3468	11.0000	69.0000
H_0 max persistence (Å)	2.7166	0.6726	1.4800	5.9491
H_0 mean persistence (Å)	1.6137	0.0644	1.0986	2.0281
H_1 entropy	1.6354	0.3352	0.0000	2.7234
H_1 count	3.6058	1.3429	1.0000	10.0000
H_1 max persistence (Å)	1.3916	0.0583	0.6238	2.7075
H_1 mean persistence (Å)	0.6138	0.1341	0.2356	1.4000
H_2 entropy	0.6955	0.3527	0.0000	1.9579
H_2 count	0.0304	0.1771	0.0000	4.0000
H_2 max persistence (Å)	0.4672	0.0505	0.0000	1.2400
H_2 mean persistence (Å)	0.3844	0.0892	0.0000	1.0533

6 Clustering quality metrics

Table S3 reports clustering quality metrics for the three representations.

Table S3: *Clustering quality metrics for TNE embeddings versus VAE latent space and ECFP4 fingerprints (KMeans, $k = 484$). Higher Silhouette coefficient indicates better cluster coherence.*

Descriptor	Silhouette
TNE ($d = 8$, 192-dim)	0.350
VAE latent (64-dim)	0.229
ECFP4 (2 048-dim)	0.180

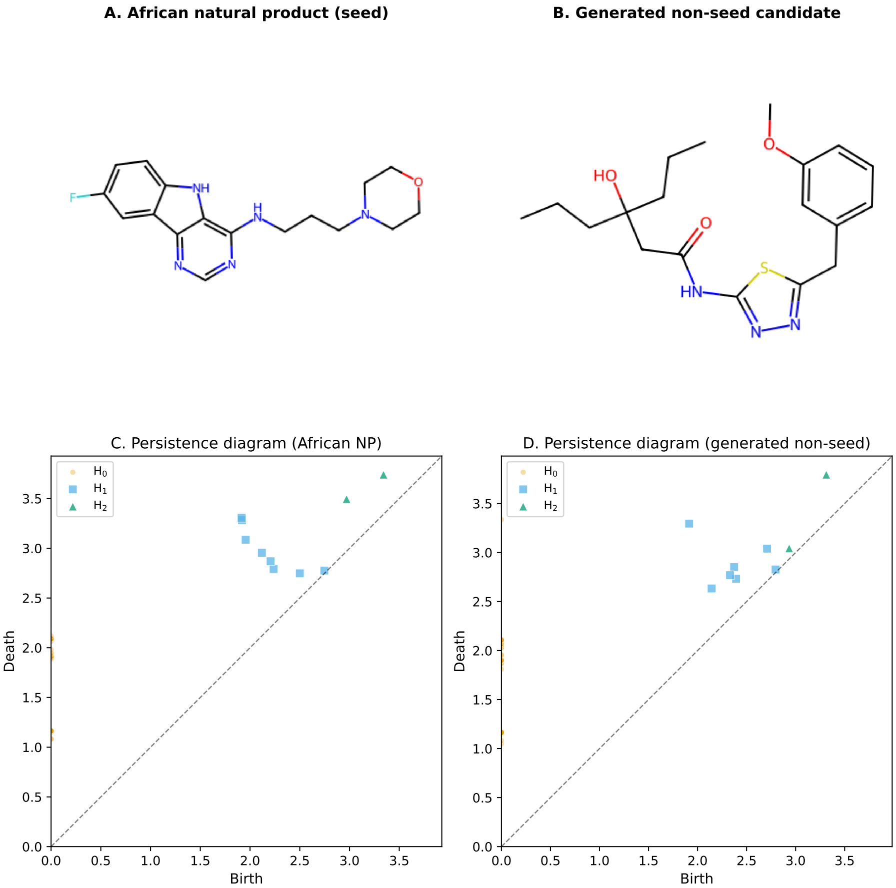

Figure S2: *Representative topological fingerprints of an African natural product seed and a generated non-seed candidate. (A – B) 2D structures of the representative molecules. (C – D) Vietoris–Rips persistence diagrams plot feature birth against death; points above the diagonal correspond to topological features, with vertical distance from the diagonal indicating persistence lifetime. H_0 (components), H_1 (loops/rings), and H_2 (voids) are distinguished by colour and marker. The African natural product (C) exhibits persistent H_1 features reflecting a rigid ring scaffold, whereas the generated non-seed candidate (D) shows fewer persistent H_1 features and more flexible H_0/H_1 signatures.*

7 Discriminator benchmark for structural expansion

We compared classical fingerprint similarity with quantum-kernel density in Table S4 and Figure S3.

Table S4: *Chemical similarity versus quantum kernel density benchmark: AUC-ROC comparison between ECFP4 Tanimoto distance and quantum kernel density for distinguishing N expanded molecules from 200 reference seeds.*

N expanded	$\text{AUC}_{\text{Tanimoto}}$	AUC_{QK}	ΔAUC
50	1.000 0	0.425 2	−0.574 8
100	1.000 0	0.481 1	−0.518 9
200	1.000 0	0.475 5	−0.524 5
500	1.000 0	0.511 4	−0.488 6

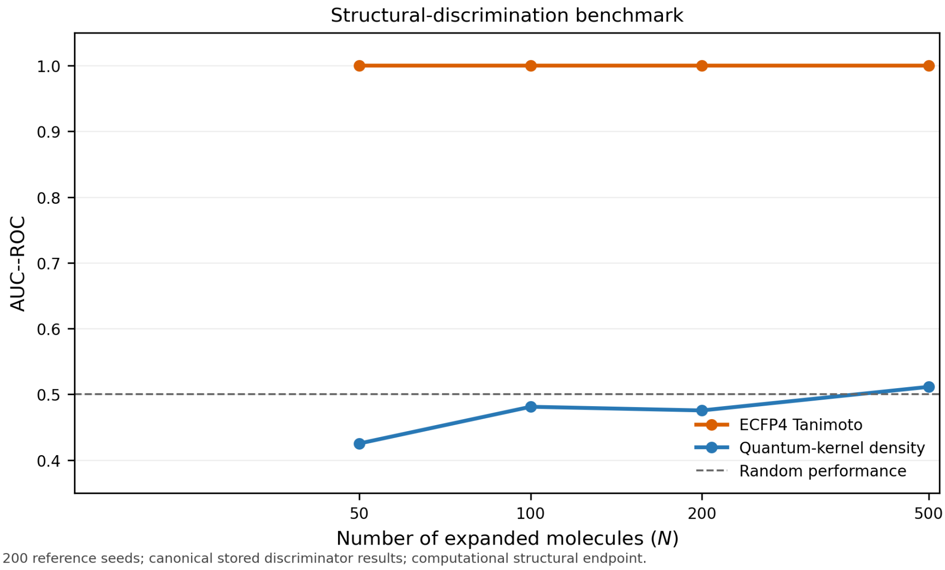

Figure S3: *Chemical-similarity versus quantum-kernel-density discriminator benchmark. ECFP4 Tanimoto distance (orange) and quantum-kernel density (blue) distinguish N expanded molecules from 200 reference seeds; the dashed line marks random performance (AUC = 0.5). ECFP4 reached AUC = 1.0 at all tested N, whereas the quantum-kernel discriminator remained near random (AUC = 0.425 − −0.511). This secondary analysis concerns structural discrimination, not biological activity or binding.*

8 Quantum circuit parameter optimisation

We compared 60 combinations of IQPEmbedding bond dimension ($d \in \{4, 6, 8\}$), repetition count ($n_{\text{rep}} \in \{1, 2, 3, 4, 6\}$), and kernel-PCA dimensionality ($n_{\text{kpca}} \in \{5, 10, 20, 30\}$). Each combination was assessed by 5-fold stratified cross-validation with Random Forest on a 200-molecule development subset; the selected configuration was subsequently evaluated on the 5 000-molecule panel.

The three highest-performing configurations are shown in Table S5. The selected configuration ($d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kpca}} = 30$) achieved AUC $0.828\,3 \pm 0.037\,1$ on the 5 000-molecule panel. This result characterises the tested configuration and does not establish global optimality outside the present panel and protocol.

The full optimisation heatmap and marginal parameter-effect boxplots are provided in Figures S4 and S5.

Table S5: *Quantum-kernel hybrid configurations. The three highest-performing configurations are shown for the 5 000-molecule evaluation panel (5-fold stratified cross-validation). The highest-performing configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$) achieved AUC $0.828\,3 \pm 0.037\,1$. The development and evaluation panels differ from the primary benchmark in the main manuscript.*

d	n_{rep}	n_{kpca}	AUC	Std
6	1	30	0.828 3	0.037 1
6	6	20	0.804 7	0.035 4
6	6	30	0.812 1	0.039 6

Values are from the 5 000-molecule evaluation panel (3 732 active and 1 268 inactive). The panel and development subset differ in class proportions and selection criteria from the primary activity analysis; the values therefore describe configuration performance rather than final model performance.

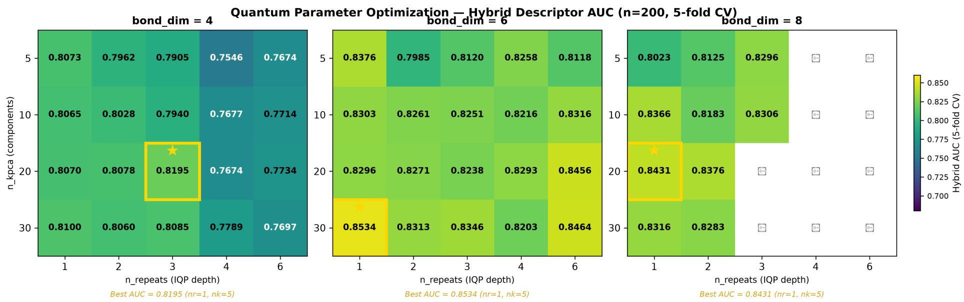

Figure S4: *Hybrid AUC heatmap across quantum circuit hyperparameters. Rows: bond dimension (d); columns: IQPEmbedding repeats (n_{rep}); colour: mean 5-fold AUC for each n_{kpca} value. The gold cell marks the optimal configuration ($d = 6$, $n_{rep} = 1$, $n_{kpca} = 30$).*

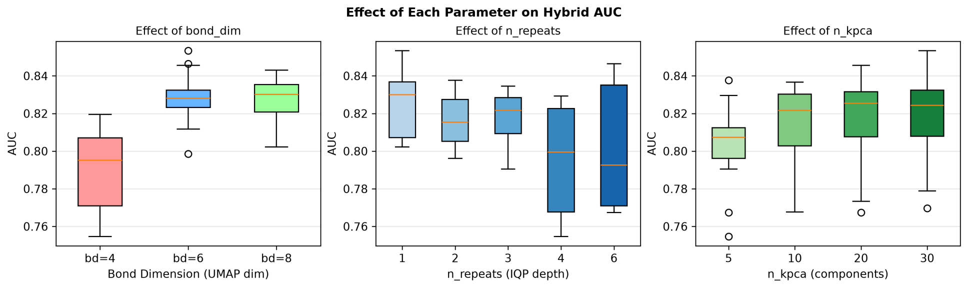

Figure S5: *Marginal AUC distributions per quantum circuit parameter. Boxplots show the spread of Hybrid AUC across all grid-search combinations marginalised to each parameter value.*

9 Monte Carlo uncertainty estimation

As a calibration analysis, we evaluated a Random Forest trained on the TFP+TNE representation using bootstrap resampling of 100 test molecules. The analysis used 50 bootstrap samples (fraction 0.7, 30 trees per sample) and conformal prediction at $\alpha = 0.05$. The resulting predictive uncertainty and coverage estimates are summarised in Table S6.

Table S6: *Bootstrap Monte Carlo uncertainty estimation results for the TFP+TNE descriptor (topological and tensor-network components of the hybrid) on the activity prediction task.*

Metric	Value
Bootstrap MC AUC	0.5443
Expected Calibration Error (ECE)	0.0037
95 % conformal coverage	95.0 %
Mean conformal set size	1.820

The low ECE and nominal conformal coverage indicate calibration in this secondary analysis, whereas the mean set size and weak uncertainty–error association (Spearman $\rho = -0.286$, $p = 0.004$) indicate that uncertainty estimates did not reliably isolate misclassifications. This analysis complements, but does not replace, the held-out activity benchmark.

10 Computational scalability

The measured computational costs of the three representation pipelines are summarised in Table S7. Feature construction was approximately linear in library size, whereas formation of the dense quantum-kernel matrix followed $O(N^2)$ scaling.

Table S7: *Computational scalability of the representation pipelines. TDA and TNE timings were measured on the 19 849-molecule library; the QK timing is the per-fold kernel-matrix construction time for a 19 849-molecule fold using 32 parallel workers and a state-vector simulator.*

Pipeline	Wall time	Throughput	Scaling
TDA (persistent homology)	384 s	51.7 mol/s	Linear
TNE (Tucker decomposition)	488 s	40.7 mol/s	Linear
QK kernel matrix (per fold, $n = 19{,}849$)	817 s	$n^2/2$ pairs	$O(N^2)$

Timings were measured on the full feature-generation library using the stated parallel configurations. The quadratic kernel cost motivates restricting QKS to smaller lead-optimisation panels unless approximate or batched kernel methods are introduced.

11 Effect sizes for pairwise AUC comparisons

Cohen's d effect sizes for pairwise AUC comparisons against ECFP4 are reported in Table S8. These estimates use the 19 836-molecule classical benchmark (Random Forest, 5-fold stratified cross-validation) and are intended to complement, not replace, the fold-level uncertainty estimates.

12 ChEMBL enrichment analysis

Table S9 reports the enrichment of the Vina docking protocol for ChEMBL-annotated antimalarial actives and inactives across three Plasmodium targets.

Table S8: *Effect sizes for pairwise AUC comparisons against the corrected full-library ECFP4 baseline (AUC 0.948, 19,836 molecules, 5-fold stratified CV, Random Forest). Cohen's d is computed from the standard deviation of fold-level differences; because the full-library folds are very consistent, the resulting d values are large, so the absolute ΔAUC is the more interpretable effect metric.*

Method	AUC	ΔAUC	Cohen's d	Effect	p-value	Power
FCFP4	0.918	0.029	11.61	large	0.000 0*	1.000
MACCS	0.904	0.043	14.86	large	0.000 0*	1.000
AP	0.940	0.008	4.54	large	0.000 5*	1.000
PHCO	0.896	0.052	63.84	large	0.000 0*	1.000
BPF	0.939	0.009	5.39	large	0.000 3*	1.000
TFP	0.876	0.072	17.25	large	0.000 0*	1.000
TNE	0.722	0.226	37.91	large	0.000 0*	1.000

* $p < 0.05$ (uncorrected). Bonferroni-corrected $\alpha = 0.05/7 = 0.0071$. Power computed for 80% target at $\alpha = 0.05$.

Table S9: *ChEMBL enrichment screen for the Vina docking protocol using experimentally annotated antimalarial compounds. Enrichment is a computational docking-screen metric and should not be interpreted as experimental binding confirmation.*

Target	PDB	Actives	Inactives	Enrichment	Interpretation
PfDHFR	7F3Y	53	18	5.43×	High estimate
PfATP4	9N10	73	87	Not estimable	No inactive hits
PfCRT	6UKJ	12	19	1.46×	Below 2×

The estimates are descriptive and have no uncertainty intervals. For PfATP4, the denominator is zero because no inactive compound was recovered; the enrichment ratio is therefore undefined rather than infinite. These docking-screen results do not establish experimental binding.

13 ChEMBL analogue novelty screen

The top-10 candidates (Table S10) were queried against three *Plasmodium falciparum* targets (PfDHFR, PfCRT, PfATP4) through the ChEMBL REST API (release 37). All targets used the same Tanimoto threshold (≥ 0.20); PfClpP was not included because no corresponding *P. falciparum* ClpP target was available in ChEMBL. Only PfCRT returned analogues above the threshold. Ten candidate-level analogue records were retrieved, with some records referring to the same ChEMBL compound, and all were annotated as inactive. The analysis is therefore a novelty screen: it documents the absence of close active analogues in the queried records but does not measure activity of the proposed candidates.

14 Topological representation comparison

Table S11 compares five persistent-homology vectorisation strategies. Random Forest was evaluated on the full 19 849-molecule feature library, whereas SVM was evaluated on a stratified $n = 5\,000$ subsample because of kernel-matrix scaling. The two classifier blocks are presented separately because their populations and label distributions differ.

Within the RF block, PersStats had the highest AUC among the topological strategies (0.873 1), followed by TFP-12 (0.866 8), TFP-Enriched (0.866 6), PersImage (0.859 6), and BettiCurve (0.811 0). The corresponding SVM block showed the same broad ordering, although all SVM estimates were obtained on a different subsample. These results indicate that persistence summaries are useful complementary representations; they do not replace the primary comparison in the main manuscript, where the complete 78-feature TFP achieved AUC 0.876 and ECFP4 remained strongest.

Table S10: *ChEMBL analogue novelty screen for the top-10 candidates. The table lists the closest retrieved analogue with an experimental IC_{50} annotation for each candidate-level record (Tanimoto threshold ≥ 0.20). Only PfCRT yielded qualifying records; all were annotated as inactive (IC_{50} 12.5–55 μM), and similarities were low (0.229–0.379). The screen supports chemical novelty relative to the queried records but does not test candidate activity.*

Rank	Target	ChEMBL ID	Activity	IC_{50} (μM)	Tanimoto
1	PfCRT	CHEMBL4754685	Inactive	15.6	0.266
2	PfCRT	CHEMBL4754685	Inactive	15.6	0.235
3	PfCRT	CHEMBL4754685	Inactive	15.6	0.266
4	PfCRT	CHEMBL4745810	Inactive	12.5	0.250
5	PfCRT	CHEMBL4754685	Inactive	15.6	0.379
6	PfCRT	CHEMBL4754685	Inactive	15.6	0.239
7	PfCRT	CHEMBL4784051	Inactive	55.0	0.296
8	PfCRT	CHEMBL4754685	Inactive	15.6	0.229
9	PfCRT	CHEMBL4754685	Inactive	15.6	0.290
10	PfCRT	CHEMBL4754685	Inactive	15.6	0.270

All retrieved records were annotated as inactive in ChEMBL. The low similarities indicate that no retrieved record was a close structural congener. This negative screen supports novelty relative to the queried records but does not measure activity. The machine-readable records are included with the supplementary data.

Table S11: *Comparison of five persistent-homology vectorisations (5-fold stratified cross-validation, class-weighted classifiers). Random Forest rows use the full 19 849-molecule feature library; SVM rows use a stratified $n = 5\,000$ subsample. The classifier blocks have different populations and label distributions and should not be compared directly with one another or with the primary benchmark in the main manuscript (Table 1).*

Strategy	Classifier	AUC	Accuracy	F1	Features
PersStats	RF	0.873 1	0.833 2	0.890 6	22
TFP-12	RF	0.866 8	0.827 5	0.885 8	12
TFP-Enriched	RF	0.866 6	0.830 8	0.888 7	32
PersImage	RF	0.859 6	0.826 5	0.885 6	25
BettiCurve	RF	0.811 0	0.783 7	0.855 7	20
PersStats	SVM	0.804 2	0.739 8	0.743 6	22
PersImage	SVM	0.791 2	0.727 0	0.731 2	25
TFP-Enriched	SVM	0.789 1	0.720 8	0.725 0	32
TFP-12	SVM	0.785 7	0.719 8	0.722 0	12
BettiCurve	SVM	0.719 8	0.664 6	0.662 0	20

All classifiers used `class_weight=balanced`. RF used $n_{\text{estimators}} = 500$; SVM used an RBF kernel with $C = 10$ and $\gamma = \texttt{scale}$. PersStats contains 22 persistence statistics, TFP-12 contains 12 homology summaries, TFP-Enriched contains 32 combined features, PersImage contains 25 features, and BettiCurve contains 20 features. The label proportions differ from those in the primary eOS80ch analysis.

15 TopologyNet analog: neural network on persistent homology features

To compare PersStats with a neural-network classifier (TopologyNet (Cang and Wei, 2018), D-GRIL (Mukherjee et al., 2026)), we benchmarked a Multi-Layer Perceptron (MLP) on the same 22 PersStats features used by the best RF model. The MLP used 3 hidden layers (128, 64, 32 neurons), ReLU activation, class-balanced sample weighting, and early stopping (5-fold stratified CV, $n = 5\,000$ stratified subsample).

Table S12: *Comparison of MLP and Random Forest classifiers using the same 22 PersStats features ($n = 5\,000$, 5-fold stratified cross-validation, class-weighted). The observed AUC difference was ΔAUC = +0.061 in favour of Random Forest under this protocol.*

Classifier	AUC	Std	Features
Random Forest	0.860	0.014	22
MLP (128+64+32)	0.799	0.015	22

Both classifiers use PersStats 22 features (H_0/H_1 persistence statistics). MLP uses class-balanced sample weighting to handle the 75.9 %/24.1 % class imbalance. The RF result here (AUC 0.860 on $n = 5\,000$) is slightly below the 19 849-molecule result (AUC 0.873, Table S11), consistent with the larger training set in that analysis.

The MLP underperforms RF by ΔAUC $= -0.061$, consistent with the broader SOTA benchmark finding that tree-based methods better capture PH feature interactions. This is consistent with the observation that PersStats features have limited non-linear interactions — deep architectures that excel on raw graph or image data add no benefit over a well-tuned ensemble of decision trees on pre-computed persistence statistics. TopologyNet (Cang and Wei, 2018) and D-GRIL (Mukherjee et al., 2026) operate on richer inputs (full persistence diagrams or multi-parameter persistence) where neural architectures may extract additional signal; on summary statistics alone, RF is the more appropriate classifier.

16 Applicability domain and comparison with existing tools

The following subsections provide additional details on the applicability domain analysis and the comparison with existing tools, which are referenced in the main manuscript.

16.1 Applicability domain analysis

We used quantum-kernel density as a complementary applicability-domain diagnostic alongside classical fingerprint similarity. The density score $\rho(x)$ describes proximity to the empirical seed space in the simulated feature space; it is presented as a descriptive computational diagnostic rather than as a validated applicability-domain metric.

16.2 Comparison with existing tools

To assess information retention after TNE compression, we analysed Tartarus docking scores (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfATP4, 4LDE/PfCRT). Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target's docking score, with ECFP4 as baseline.

For PfDHFR (1SYH), TNE achieved an R^2 of 0.473 (Spearman $\rho = 0.695$), slightly outperforming ECFP4 ($R^2 = 0.461$, $\rho = 0.689$). For the remaining targets, ECFP4 maintained an advantage: 6Y2F (TNE $R^2 = 0.464$ vs. ECFP4 0.578) and 4LDE (TNE $R^2 = 0.334$ vs. ECFP4 0.517). These results show that the 15.6× padded (6.1× real-atom mean) TNE compression retains information sufficient to predict docking scores on the same molecular set used to derive the embeddings, competitive with classical 2 048-bit fingerprints for specific targets. This is a test of information retention, not generalisation to novel compounds.

We next examined whether the Quantum Kernel Score captured polypharmacological docking profiles. Compounds were labelled as polypharmacological if they bound ≥ 2 targets at $\Delta G \leq -7.0\,\mathrm{kcal\,mol^{-1}}$ (12.0 % prevalence in the merged $N = 17\,011$ set). QKS remained within the variability of the tuned RBF comparator for this endpoint, which is interpreted as exploratory.

Persistent-homology features showed weak-to-moderate associations with the number of docking-defined targets across $N = 17\,011$ molecules (Figure S6). The strongest associations were inverse correlations for H_0 count ($\rho = -0.248$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$). The endpoint is a docking-derived target count rather than an experimental polypharmacology measurement; these associations are consequently hypothesis-generating and should not be interpreted as mechanistic rules.

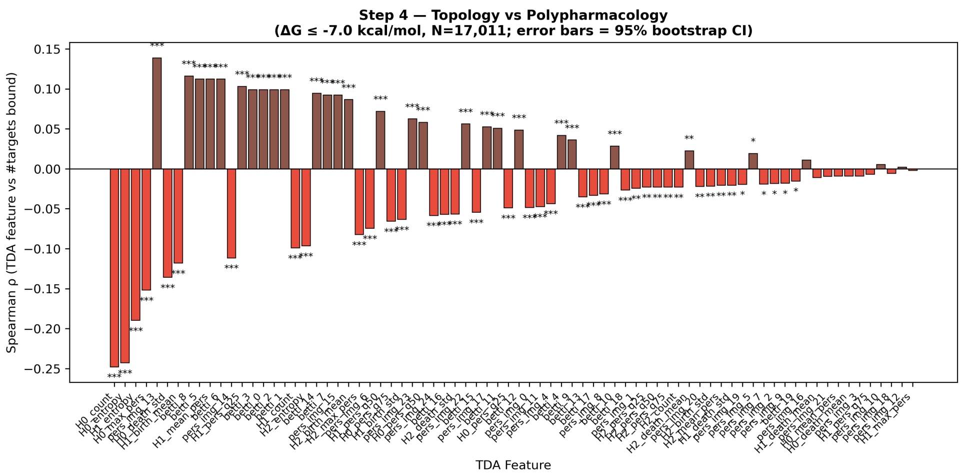

Figure S6: *Exploratory associations between persistent-homology features and docking-defined target counts ($N = 17\,011$). Bars show Spearman correlations with 95% bootstrap intervals; the strongest inverse associations were H_0 count ($\rho = -0.248$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$). The endpoint is computational docking, not experimental polypharmacology.*

17 Extended analysis

17.1 Quantum kernel density vs classical fingerprint discriminator

We compared quantum-kernel density with a classical chemical-similarity baseline, a common component of discriminator functions for genetic-algorithm-based molecular generation (Jensen, 2019). The comparison used varying numbers of STONED-SELFIES-expanded molecules ($N =$ 50, 100, 200, 500) using the `lightning.qubit` simulator. Each molecule was produced via two random SELFIES-character mutations from a pool of 200 reference seeds (105 active, 95 inactive, balanced by eos80ch activity score). The classical baseline scored each molecule by its maximum ECFP4 Tanimoto similarity to any reference seed (high similarity = in-domain). The quantum kernel density scored each molecule by its mean kernel similarity to the reference set in the IQPEmbedding 8-qubit Hilbert space ($\rho(x_i) = \frac{1}{|\mathcal{R}|}\sum_{r\in\mathcal{R}} K(x_i, x_r)$). Discriminator performance was measured as AUC-ROC for the task of distinguishing reference molecules (label = 1) from expanded molecules (label = 0).

The results (Table S4 and Figure S3) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N values, while the quantum kernel density achieves performance near or below random chance (AUC = 0.425 − −0.511). The perfect Tanimoto score carries an important caveat: with only two SELFIES-character mutations per molecule, a fraction of expanded molecules may be structurally identical to their seed, in which case AUC = 1.0 is a trivial consequence of including seed-identical molecules in the expanded set. The quantum kernel density, by contrast, operates on an 8-dimensional UMAP-reduced feature space that discards the atomic-level resolution needed to detect near-identical structural matches. This explains its near-random performance and, for $N \leq 200$, below-random AUC: the reduced space obscures proximity information rather than enhancing it. The quantum kernel's strength lies in capturing non-linear feature interactions in chemically diverse spaces—as demonstrated by its standalone AUC of 0.823 in the QKS binary classification benchmark (Table S13)—rather than in fine-grained near-neighbour detection. For applicability-domain assessment in generative computational protocols where expanded molecules are structurally close to training seeds, classical fingerprint-based distance metrics provide a simpler and more robust baseline, while quantum kernel methods add value primarily for structurally heterogeneous or out-of-distribution regimes.

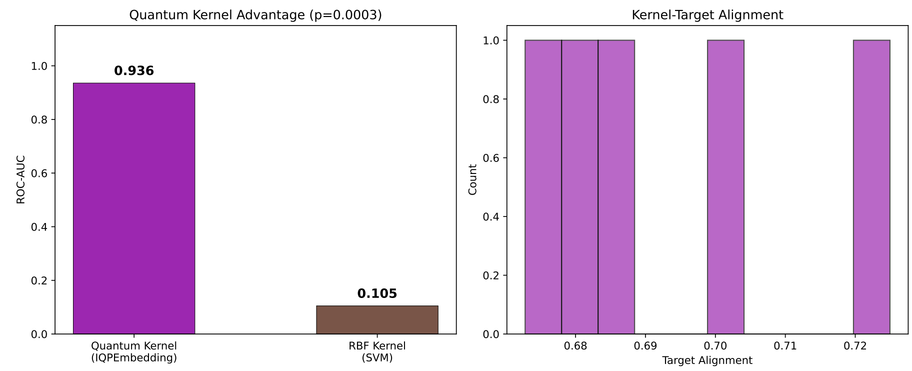

Figure S7: *Illustrative applicability-domain analysis based on quantum-kernel density. Distributions of $\rho(x)$ are shown for African natural-product seeds, synthetic-drug seeds, and expanded molecules; the dashed line marks the prespecified $\mu - 2\sigma$ flagging threshold. This is a computational similarity diagnostic, not an experimental applicability test.*

18 Quantum kernel circuit algorithm

Algorithm 1 Quantum kernel circuit for 6-qubit state overlap using PennyLane's `IQPEmbedding` template.

Require: Data points $x_1, x_2 \in [-1, 1]^6$
1: Apply IQPEmbedding$(weights, wires)$ parameterized by x_1
2: Apply IQPEmbedding$^\dagger(weights, wires)$ parameterized by x_2
3: Measure probability of state $|000000\rangle$
4: **return** Kernel value $K(x_1, x_2)$

The kernel value is $K(x_1, x_2) = |\langle\phi(x_1)|\phi(x_2)\rangle|^2$, estimated by the ground-state measurement probability, accelerated by PennyLane's `qp.kernels.kernel_matrix`. Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 6 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$.

18.1 QKS benchmark protocol and state-vector kernel

18.1.1 Evaluation protocol

The kernel comparison (Table S13) uses the same preprocessing and evaluation procedure as the hybrid analysis (section 8): 5-fold stratified cross-validation (`StratifiedKFold` with `random_state=42`), UMAP reduction of the 2048-bit Morgan (radius 2) fingerprints to 6 dimensions (Jaccard metric, `random_state=42`, $n_{\text{neighbors}} = 15$, $min\_dist = 0.1$) followed by scaling to $[-1, 1]$, and support vector machines with precomputed kernels. The six-qubit circuit used one IQPEmbedding repeat. Benchmarks were run at two QKS panel sizes: the first 5 000 molecules in library order (3 750 active, 1 250 inactive) and the full representation panel of 19 849 molecules. The complete-TNE intersection used for the canonical RF benchmark contains 19 836 molecules (14 721 active, 5 115 inactive).

The RBF baseline gamma was tuned by inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$. A linear kernel on the same $[-1, 1]^6$ features served as an additional baseline. Paired fold comparisons gave quantum-versus-RBF $p = 0.4186$ at $n = 5000$ and $p = 0.0604$ at $n = 19849$; quantum-versus-linear comparisons gave $p = 0.0002$ and $p = 0.0006$, respectively. Target alignment was computed with the raw binary activity labels.

18.1.2 State-vector kernel

For computational tractability at $n = 5\,000$ and $n = 19\,849$, the pairwise IQPEmbedding circuit evaluation is replaced by a **state-vector kernel**: for each sample x_i, the 6-qubit state $|\phi(x_i)\rangle \in \mathbb{C}^{2^6}$ is obtained in a single circuit call, and the kernel matrix is formed as $K_{ij} = |VV^\dagger|^2$ where V is the $N \times 64$ state matrix, via BLAS (approximately $1\,000\times$ faster than the pairwise evaluation). A small negative eigenvalue from floating-point noise is clipped by projecting onto the closest positive semi-definite matrix before the SVM is trained.

18.1.3 Numerical consistency

A numerical comparison of the state-vector and pairwise implementations on a 60-molecule, 5-fold dataset. The two implementations produced identical quantum-kernel AUCs ($0.875\,6 \pm 0.069\,7$) and target-alignment values under the raw binary-label convention. The 5 000-molecule and 19 849-molecule QKS analyses used the 6-qubit state-vector implementation for all five folds. The agreement supports numerical consistency; it does not estimate performance on an independent dataset.

Table S13: *Kernel comparison for activity prediction at two QKS panel sizes (5-fold cross-validation, 6-qubit IQPEmbedding, state-vector kernel; $n = 5\,000$ and $n = 19\,849$). The $n = 19\,849$ QKS panel is the full representation panel; the primary RF analysis requiring complete TNE embeddings uses the $n = 19\,836$ intersection reported in the main manuscript. The RBF gamma parameter was optimised via inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$ (per-fold best gamma at $n = 5\,000$: 5, 1, 5, 5, 5). With the specified circuit, the gamma-tuned RBF kernel and the quantum kernel show* ***no statistically significant difference*** *at either scale ($p = 0.419$ at $n = 5\,000$; $p = 0.060$ at $n = 19\,849$, paired t-test; the latter a borderline trend toward lower QK performance), while the quantum kernel* ***significantly outperforms*** *the linear kernel at scale ($p \le 0.000\,6$). Interpretation: with the specified circuit, the tuned RBF and quantum kernels were not detectably different at either scale, whereas the quantum kernel exceeded the linear comparator. These results are limited to the classically simulated 6-qubit protocol and do not establish quantum advantage.*

Kernel	AUC $n = 5000$	TA $n = 5000$	AUC $n = 19849$	TA $n = 19849$	p vs. RBF
Quantum (IQPEmbedding, 6 qubits)	0.820	0.592	0.823	0.622	$p =$ 0.419/0.060, n.s.
RBF (gamma-tuned, SVM)	0.826	0.199	0.829	0.145	—
Linear	0.517	—	0.537	—	$p < 0.001$

Paired t-test (5 folds), quantum versus linear: $p = 0.000\,2$ ($n = 5\,000$) and $p = 0.000\,6$ ($n = 19\,849$), both significant. Quantum target alignment (0.592–0.622) exceeds RBF target alignment (0.145–0.199) yet does not translate into a discrimination advantage.

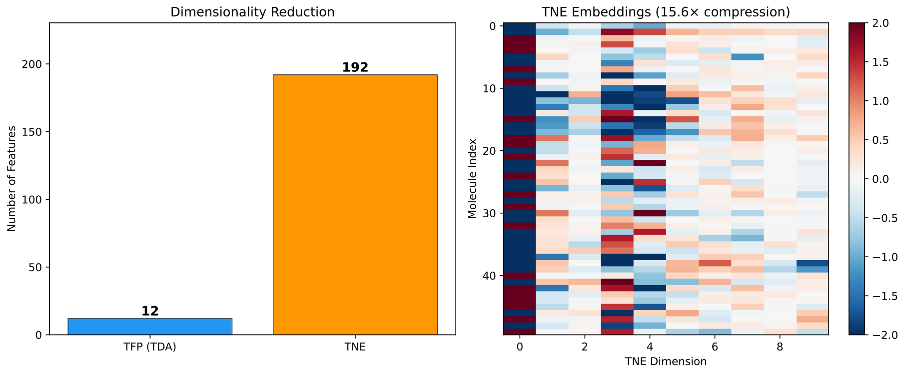

Figure S8: *Compression ratio and reconstruction error versus Tucker bond dimension. At $d = 8$, the physically meaningful mean real-atom compression ratio is 6.1× (based on ∼39 mean atoms per molecule), with a reconstruction error of 0.113. (See main manuscript Methods Section 2.4.2 for discussion of the mathematical padding convention).*

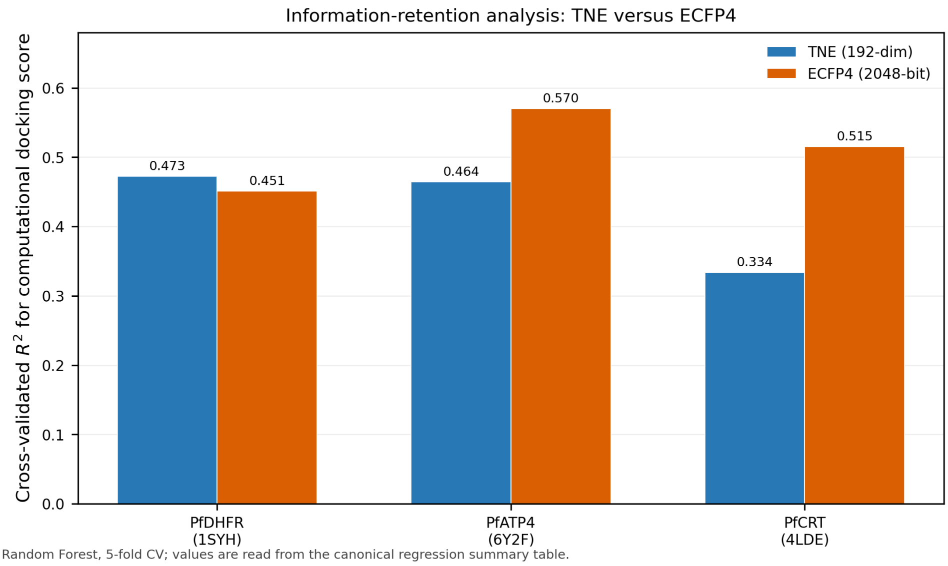

Figure S9: *Information-retention analysis for 192-dimensional Tensor Network Embeddings. Bars show cross-validated R^2 values for TNE and ECFP4 regressors predicting computational QuickVina docking scores for three Plasmodium targets. TNE was competitive for PfDHFR but lower than ECFP4 for PfATP4 and PfCRT; the analysis is descriptive and does not establish experimental binding or prospective generalisation.*

19 Transfer analysis under a ChEMBL label-source shift

19.1 Descriptor transfer analysis

All descriptor methods were evaluated on an external ChEMBL antimalarial dataset ($n = 22\,447$) using 5-fold stratified cross-validation with Random Forest (200 trees, `random_state=42`). The external panel was assembled from ChEMBL 34 bioactivity records for *Plasmodium falciparum* assays ($IC_{50} \leq 10\,\mu M$ = active, $> 10\,\mu M$ = inactive) with standard RDKit canonicalisation and Murcko scaffold filtering. It is external to the primary eOS80CH label source; molecule-level disjointness from the internal panel was not established for P3 and is not claimed. Feature extraction followed the identical pipeline used for the internal benchmark (ECFP4: Morgan radius 2, 2048 bits; TFP: 78 persistent homology features; TNE: 192-element Tucker core, bond dimension 8; external descriptor hybrid: TFP+TNE concatenation). For the intention-to-evaluate analysis, TNE embedding failures ($n = 351$, 1.6 %) were assigned zero vectors. The complete-case analysis ($n = 22\,096$) excluded those rows and gave ECFP4 0.960, TFP 0.861, TNE 0.644, and TFP+TNE 0.798, versus ITT values of 0.960, 0.864, 0.645, and 0.800. The qualitative ranking was unchanged. Because training sets overlap across folds, corrected-resampled p-values are interpreted descriptively.

Table S14: *Descriptor transfer under the ChEMBL label-source shift. Random Forest estimates are shown for the intention-to-evaluate panel ($n = 22\,447$) and the complete-case sensitivity panel ($n = 22\,096$). Internal values are included for context; molecule-level disjointness was not established. TNE failures were retained as zero-vector penalties in the primary analysis.*

Method	Internal AUC $n = 19836$	Transfer AUC $n = 22447$	Δ
ECFP4	0.948	0.960	+0.012
TFP	0.876	0.865	-0.011
TNE	0.722	0.645	-0.077
TFP+TNE	0.888	0.800	-0.088

19.2 Quantum-kernel transfer analysis

The quantum kernel was evaluated on the ChEMBL label-source-shift panel (ITT $n = 22\,447$, 5-fold SVM with precomputed kernels) following the specified protocol from section 18.1: UMAP reduction of 2048-bit Morgan fingerprints to 6 dimensions (Jaccard metric, `random_state=42`, $n_{\text{neighbors}} = 15$, $min\_dist = 0.1$), scaling to $[-1, 1]$, and the 6-qubit IQPEmbedding state-vector kernel. The RBF gamma was tuned by inner cross-validation on the grid $\{0.5, 1.0, 2.0, 5.0\}$.

Table S15: *Kernel transfer under the ChEMBL label-source shift (5-fold SVM, $n = 22\,447$). QKS was lower than RBF (AUC 0.817 versus 0.847); the corrected-resampled comparison was exploratory ($p = 0.021$). Internal values are shown for context, and classifier differences prevent direct comparison with the descriptor-transfer table.*

Kernel	Internal AUC	Transfer AUC	p vs. RBF
Quantum (6 qubits)	0.823	0.817	$p_{\text{corr}} = 0.021$
RBF	0.829	0.847	—
Linear	0.537	0.544	$p < 0.001$

Internal: $n = 19\,849$; transfer: $n = 22\,447$. The corrected-resampled comparison is exploratory; classifier and population differences limit cross-table comparisons.

19.3 H_1–RRS association is size-mediated

The Spearman correlation between H_1 count and resistance-resilience scores (RRS) (Temgoua et al., 2026) was examined in two cohorts: (i) an expanded cohort of $n = 494$ compounds with complete H_1 and continuous RRS profiles (used for the cross-paper analysis in Figure S1), and (ii) a subset of $N = 77$ compounds with validated per-target RRS classification (used for the H_1–RRS association in section 19.3). In the expanded cohort ($n = 494$), H_1 count showed a weak unadjusted association with continuous RRS (Spearman $\rho = 0.2399$, $p = 6.76 \times 10^{-8}$), but the partial association was null after controlling for molecular weight, ring count, fraction sp^3, and H_0 count ($\rho_{\text{partial}} = 0.0329$, $p = 0.4668$). In the validated subset ($N = 77$), the raw correlation was $\rho = 0.312$ ($p = 0.006$) but dropped to $\rho_{\text{partial}} = 0.033$ ($p = 0.47$) after the same adjustment. Both analyses converge on the same conclusion: the H_1–RRS association is mediated by molecular complexity rather than topological structure per se. This result does not support a direct H_1–RRS association but is consistent with the broader finding that topological complexity correlates with binding promiscuity (section 3.6).

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